Wenjia Ge

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Research Interests

Semiconductor and optoelectronic materials and devices; thin-film fabrication and characterization; optical measurement and sensing; quantum devices.

Education

New York University, Tandon School of Engineering *Sept 2024 – Dec 2026 (expected)*
New York, NY
M.S., Electrical Engineering, GPA: 3.926/4.0 | GRE: 327
Relevant coursework: Physics of Quantum Computing, Physics of Nanoelectronics, Electromagnetic Theory and Applications, Digital Signal Processing, Real-Time Embedded Systems, Applied Matrix Theory, Probability and Stochastic Processes

Shenzhen University, Shenzhen, China *Sept 2020 – July 2024*
B.S., Physics, GPA: 3.65/4.0 | Outstanding Undergraduate Thesis Award
Relevant coursework: Quantum Mechanics, Electrodynamics, Thermodynamics and Statistical Physics, Solid-State Physics, Optics, Information Photoelectronics, Thin Film Physics and Technology, Nuclear Physics, Modern Physics Experiment

Publications

Ge, W., Wen, P., Li, Z., Luo, Q., Zhao, C., Wang, J., & Yang, M. (2023). Application of the Method of Manufactured Solutions to Verify a Nuclear Reactor Physics Computational Code Using the Method of Characteristics. *Proceedings of the 30th International Conference on Nuclear Engineering (ICONE30)*, Paper 1702. doi:10.1299/jsmeicone.2023.30.1702

Contributor, *Wanderland*: a multi-sensor capture and 3D-reconstruction pipeline for embodied-AI simulation (CVPR 2026 Highlight), NYU AI4CE Lab.

Research Experience

Undergraduate Thesis Research: Sb_2S_3 Thin-Film Solar Cells *2024*
Shenzhen University (Advisor: Dr. Guangxin Liang), Shenzhen, China

- Fabricated FTO/CdS/Sb_2S_3/Spiro-OMeTAD/Au thin-film solar cells using chemical bath deposition and studied a thioacetamide surface treatment to improve absorber-layer and interface properties.
- Characterized crystallinity, surface morphology, wettability, and photovoltaic performance using XRD, SEM, contact-angle measurement, J–V testing, and monochromator-based EQE, comparing treated devices with control samples.
- Extracted band gap and Urbach energy from EQE (27 to 22 meV) and ideality factor from dark J–V (3.51 to 2.44), consistent with reduced defect-related recombination.
- Improved power conversion efficiency from 4.43% to 5.43% through surface and interface modification; received the Outstanding Undergraduate Thesis Award.

Undergraduate Research: Nuclear Reactor Physics *2020 – 2024*
Shenzhen University, Shenzhen, China

- Applied the Method of Manufactured Solutions to verify the spatial-discretization accuracy of the Method of Characteristics in nuclear reactor physics simulation; carried out numerical modeling, coding, debugging, and verification.
- Resulted in a first-author publication at ICONE30.

Research Assistant: Tactile Sensing and Surgical Navigation *Feb – Aug 2026*
New York University (PI: Prof. Rui Li), New York, NY

- Built an Arduino and Python (pyserial) data-acquisition workflow to stream, plot, and analyze two tactile sensors in real time.
- Characterized baseline noise and contact response, tuned detection thresholds from measured ADC ranges, and quantified cross-talk between channels during simultaneous operation.
- Evaluated waterproof encapsulation (PDMS, Ecoflex, Parylene C) and defined bare-vs-coated metrics for sensitivity, baseline stability, noise, and response time.
- Built a concentric-tube surgical robot simulation in vascular anatomy, integrating a haptic device with ROS2, RViz, 3D Slicer, and VMTK centerline models.

Research Assistant, AI4CE Lab (PI: Prof. Chen Feng) *Apr 2026 – Present*
New York University, New York, NY

- Capture real indoor and outdoor scenes with MetaCam (LiDAR and fisheye cameras) and process them into point-cloud datasets and 3D Gaussian Splatting reconstructions; build HPC workflows for preprocessing and quality control.
- Serve as student team lead for Wanderland (CVPR 2026 Highlight), training new members on capture protocols.

Research Intern, Embodied AI and Robotics, AI2 Robotics *Oct 2025 – Jan 2026*
Shenzhen, China

- Designed test protocols and performance metrics for seven robot subsystems, including perception, localization, motion control, and safety; documented failure cases for cross-team troubleshooting.

Teaching Experience

Teaching Assistant, ECE 6113 / BE 6403 Digital Signal Processing I *Fall 2026*
NYU Tandon School of Engineering, New York, NY

- Hold weekly office hours (about 2 hours per week) and answer students' questions on Digital Signal Processing concepts and course material.
- Write midterm and final exam questions and grade student work.
- Coordinate weekly synchronous sessions, lecture recordings, and course materials on Brightspace.

Graduate Student Mentor *Summer 2026*
NYU Tandon School of Engineering, New York, NY

- Advise incoming graduate students on course selection, academic planning, and campus and career resources.

Other Experience

Hardware Test Engineer Intern, Shenzhen Zhi'ao Technology Co., Ltd. *July – Aug 2023*
Shenzhen, China

- Tested embedded motherboards and industrial PCs for thermal throttling, memory compatibility, and 24/48-hour load stability; wrote C and Python test scripts for serial, Ethernet, and GPIO interfaces; contributed to a 15% reduction in hardware failure rates.

Honors and Awards

- Mentor Scholarship, NYU Tandon School of Engineering, Fall 2026
- Violet Internship & Research Award (VISAA), New York University
- Outstanding Undergraduate Thesis Award, Shenzhen University

Skills

Experimental: chemical bath deposition, spin coating, annealing, thermal evaporation, glovebox work, XRD, SEM, contact-angle measurement, J–V, monochromator/EQE, oscilloscope, ADC data acquisition, noise analysis
Computation: Python (NumPy, pandas, matplotlib), MATLAB, C, C++, Verilog, Linux, Git, HPC workflows, numerical verification
Embedded and 3D: STM32, Arduino, ROS2, point-cloud processing, 3D Slicer, VMTK
Languages: Mandarin Chinese (Native), English (Fluent)